Grid-Forming Control of a PMSG Wind Turbine Using Rotor Kinetic-Energy Reserve for Inertia and Fast Frequency Support in Weak Grids

Step 1 Progress Report (Mid-Semester Evaluation), Wind Electrical Systems Design Project

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Abstract

Converter-interfaced wind turbines add no natural inertia, and grid-following (GFL) converters that synchronise through a phase-locked loop lose stability as the short-circuit ratio (SCR) falls. This report presents Step 1 of a project on a grid-forming (GFM) PMSG wind turbine that holds a 10 percent power reserve by running on the overspeed side of the C_p-optimal line, so that rotor kinetic energy and reserve are released together. We built an aerodynamic model ($C_{p,max} = 0.480$ at $\lambda_{opt} = 8.10$, below the Betz limit 0.593), a frequency-dynamics model with 40 percent wind, and a quasi-static weak-grid eigenvalue study; all 16 verification gates pass. After a 50 MW load step the GFM plant lifts the nadir from 49.659 Hz (GFL, MPPT) to 49.906 Hz, inside the IEGC band, halves the instantaneous RoCoF (0.162 against 0.332 Hz/s) and raises system inertia from 3.76 s to 8.51 s. The reserve, not the control form, sets the nadir: GFL with the same reserve reaches the same value. The decisive GFM advantage is weak-grid stability: GFL loses stability at SCR 1.30 to 1.37, GFM stays stable to the static limit near 1.0. Step 2 adds detailed dq converter models, current limiting and realistic frequency-measurement windows.

Key numbers.

Quantity	Result
Nadir after the 50 MW step	49.659 Hz (C1, GFL with MPPT) against 49.906 Hz (C4, GFM with 10% reserve), inside the IEGC band 49.90 to 50.05 Hz
RoCoF at the step (instantaneous)	0.332 Hz/s (C1) against 0.162 Hz/s (C4); GFL with the same reserve (C3) stays at 0.332 Hz/s
Late sag of C2 (support, no reserve)	49.815 Hz at 11.2 s, settling at −0.160 Hz: worse than no support (C1, −0.148 Hz)
System inertia H_{sys}	3.76 s (C1, GFL) against 8.51 s (C4, GFM); all-synchronous C0: 5.88 s
Reserve and kinetic energy per turbine at 9 m/s	reserve 84.4 kW; releasable 1.357 MJ down to the MPPT speed (3.232 MJ down to 0.5 pu speed)
Weak-grid stability	GFL critical SCR 1.30 to 1.37 (PLL 40 to 10 Hz); GFM stable down to the static limit near 1.0 (last equilibrium SCR 1.01)
Nadir versus wind share, 10 to 60%	GFL-MPPT worsens from 49.751 to 49.547 Hz; GFM improves from 49.811 to 49.928 Hz

1 Motivation and problem

Converter-interfaced wind turbines give no natural inertia: the rotor's kinetic energy is hidden behind the power-electronic interface. Conventional grid-following (GFL) control, which synchronises through a phase-locked loop (PLL), also weakens as the short-circuit ratio (SCR) falls, because the PLL must track a voltage that the converter itself disturbs.

The Indian context makes this urgent. About 54 GW of wind was installed by late 2025. A GRID-INDIA discussion paper on grid-forming inverters (February 2026) recommends GFM but focuses on battery storage and solar, and says little specific about wind [11]; this project addresses that gap. Ember reports 68 events between January 2022 and July 2025 in which more than 1 GW of renewable generation was lost in a single incident, traced to ride-through failures, protection settings and PLL instability under weak-grid conditions, concentrated in the Rajasthan RE complexes [12]. The CEA technical-standards amendment of 2026 applies from 1 April 2027 [13].

2 Link to course theory

From the course notes [9]: the Betz limit $C_p \leq 16/27 \approx 0.593$; the power coefficient $C_p(\lambda, \beta)$ versus tip-speed ratio (TSR) λ; and operation on the C_p-optimal line (MPPT), $P = K_{opt}\omega^3$. The control strategies covered (pitch control, electrical power control through the power-electronic interface, yaw, and braking) map directly onto this work: pitch limits speed, the converter shapes electrical power, while yaw and braking are not needed for the frequency problem.

Key idea. Deliberately operate *off* the C_p-optimal line, on the overspeed side. There, slowing the rotor *raises* aerodynamic power, so kinetic energy and the power reserve are released together and no secondary dip follows (unlike MPPT-based inertia emulation, which recovers by drawing power back).

3 Objectives and scope

The full project has three steps. **Step 1 (this report):** aggregated aerodynamic and swing models, GFL versus GFM frequency support on a Kundur-type test system, and a quasi-static weak-grid stability study. **Step 2:** detailed PMSG dq model with converters and DC link, dual-port GFM, current limiting and ride-through. **Step 3:** IEEE 9-bus test, turbulence, comparison with Indian grid-code requirements and the final report (Section 7).

4 Models and methods

4.1 Aerodynamics

The Heier C_p model [8] is

$$C_p(\lambda, \beta) = c_1\Big(\frac{c_2}{\lambda_i} - c_3\beta - c_4\Big)e^{-c_5/\lambda_i} + c_6\lambda, \quad \frac{1}{\lambda_i} = \frac{1}{\lambda + 0.08\beta} - \frac{0.035}{\beta^3 + 1}, \tag{1}$$

with $(c_1, \ldots, c_6) = (0.5176, 116, 0.4, 5, 21, 0.0068)$. Turbine data are in Table 1.

Table 1: Turbine data.

Rotor radius R	35.4 m	Inertia constant H	3 s
Rating	2 MW	λ_{opt}	8.10
Rated wind speed	12 m/s	$C_{p,max}$	0.480
Rated speed	26.2 rpm	Speed limits	0.5 to 1.2 pu

4.2 MPPT, deloading and rotor swing

MPPT follows $P = K_{opt}\omega^3$. Overspeed deloading selects the operating TSR $\lambda_{del} > \lambda_{opt}$ such that

$$C_p(\lambda_{del}, 0) = 0.9\, C_{p,max}, \tag{2}$$

with pitch used only above the rated wind speed (rated speed held, power capped) or where the 1.2 pu speed limit binds. The rotor obeys

$$2H\,\omega\,\frac{d\omega}{dt} = P_{aero} - P_e. \tag{3}$$

4.3 Frequency-support laws

The five cases of Table 2 differ only in the wind-plant law. In the grid-following cases C2 and C3 the converter adds synthetic inertia and droop, computed from the PLL-measured frequency ω_m (per unit, first-order lag of 50 ms), to its power set-point:

$$\Delta P_{SI} = -2H_v\frac{d\omega_m}{dt}, \qquad \Delta P_{dr} = -\frac{\omega_m - 1}{R_w}, \tag{4}$$

with $H_v = 5$ s and $R_w = 0.05$, both terms clipped. In C2 (no reserve) the set-point is the MPPT law $K_{opt}\omega_r^3$. In C3 (10 percent reserve) it is the constant deloaded set-point

$$P_e = P_{del} - g(\omega_r) + \Delta P_{SI} + \Delta P_{dr}, \qquad g(\omega_r) = K_g \max(0, \omega_{opt} - \omega_r), \tag{5}$$

where P_{del} is the constant deloaded set-point (759.4 kW per turbine at 9 m/s) and g is a stall guard with $K_g = 5$ pu that acts only if the rotor passes the C_p peak. The GFM case C4 is a virtual synchronous machine (VSM) with the same constant set-point [1, 4]:

$$2H_v\frac{d\omega_v}{dt} = P_{del} - g(\omega_r) + \Delta P_{dr} - P_e - D_d(\omega_v - \omega_3), \tag{6}$$

where $\Delta P_{dr} = \mathrm{clip}\big(-(\omega_v - 1)/R_w, \pm\Delta P_{res}\big)$ is the droop term clipped at the reserve ($\Delta P_{res} = 84.4$ kW per turbine, 40 MW for the plant) and ω_3 is the PCC (load-bus) frequency used for the damping, as in D'Arco and Suul [4]. The damping gain $D_d = 100$ was chosen from the ladder 40, 60, 80, 100 to give the electromechanical mode of the SG fleet and the VSM a damping ratio of 0.312; referencing the damping to the washed-out own frequency instead reached at most 0.153. A constant set-point is used (and not an MPPT-type law $k_{del}\omega_r^3$) because on the overspeed side $dP_{aero}/d\omega < 0$, so a constant set-point is self-stabilising, and as the rotor slows the aerodynamic power rises and the full reserve is released.

4.4 Test system and cases

An equivalent synchronous-generator (SG) fleet ($H = 5$ s) with a reheat steam governor [7] (droop 0.05, $T_G = 0.2$ s, $T_{CH} = 0.3$ s, $F_{HP} = 0.3$, $T_{RH} = 7$ s) is connected through a DC power-flow network with $X_{13} = 0.25$ and $X_{23} = 0.20$ pu on 1000 MVA. Wind supplies 40 percent of the load at 9 m/s (474 turbines of 2 MW, 948 MW installed); the disturbance is a 50 MW (5 percent) load step at $t = 1$ s.

Table 2: Simulated cases (wind-plant law).

Case	Wind-plant control
C0	All-synchronous baseline (no wind)
C1	Grid-following, MPPT $P_e = K_{opt}\omega_r^3$, no frequency support
C2	Grid-following MPPT with synthetic inertia and droop, no reserve
C3	Grid-following with synthetic inertia and droop and a 10 percent overspeed reserve
C4	Grid-forming VSM with the same 10 percent reserve (proposed)

4.5 Weak-grid model

A quasi-static small-signal model compares an SRF-PLL GFL converter against a VSM GFM converter while sweeping the SCR [5, 15].

4.6 Assumptions

The wind plant is aggregated into one equivalent turbine; the converter is averaged (no switching); network flows use DC power flow; and the network is quasi-static (no electromagnetic transients). Results are therefore valid for electromechanical time scales only.

5 Results

5.1 Aerodynamic design

Aerodynamics and the Betz limit. Fig. 1 shows the Heier $C_p(\lambda, \beta)$ curves for $\beta = 0$ to 20°. At $\beta = 0$ the coefficient peaks at $C_{p,max} = 0.480$ at $\lambda_{opt} = 8.10$, which is 81.0 percent of the Betz limit $16/27 = 0.593$. The gap is the part of the wind power that real blades lose to drag, tip losses and wake rotation, none of which the actuator-disc argument contains. The turbine reaches its rated 2 MW at 12.00 m/s and 26.22 rpm (2.746 rad/s); the optimal-torque constant is $K_{opt} = 96\,616$ W/(rad/s)^3 and the rotor inertia is $J = 1.592 \times 10^6$ kg m^2 ($H = 3$ s). Below rated wind speed the converter realises the C_p-optimal line by controlling the electrical power as $P_e = K_{opt}\omega^3$ (Fig. 2, dashed line). This is electrical power control through the converter, and it keeps the rotor on top of the C_p hill. A rotor parked there has nothing to give: releasing its kinetic energy slows it to the left of the peak, where C_p and aerodynamic power fall, so the support must be repaid afterwards. This is the origin of the secondary dip met in Task B.

Deloading, pitch and the reserve. For a 10 percent reserve the converter instead holds the rotor to the right of the peak, at $C_p(\lambda_{del}) = 0.9\,C_{p,max} = 0.432$. At 9 m/s this means $\omega_{del} = 23.28$ rpm against $\omega_{opt} = 19.67$ rpm ($\lambda_{del} = 9.59$ against 8.10) and $P_{del} = 759.4$ kW against 843.8 kW available, a reserve of 84.4 kW. The worked example $P = \frac{1}{2}\rho A v^3 C_p$ with $\rho = 1.225$ kg/m^3, $A = 3937$ m^2 ($R = 35.4$ m) reproduces the 0.844 MW. On the overspeed side of the peak (Fig. 2, green line) $dP_{aero}/d\omega < 0$: if the converter draws more than the rotor receives, the rotor decelerates, λ moves back toward λ_{opt} and C_p and P_{aero} rise. This is the physical basis of the project: kinetic energy and reserve are released together. Overspeed is used from 6 to 11 m/s. From the rated wind speed of 12 m/s, where the available power is capped at 2 MW, the rotor is held at its rated speed and the reserve is created by pitch control instead: $\beta = 1.18°$, 4.16° and 8.48° at 12, 13 and 14 m/s, holding 1.8 MW and keeping a 200 kW reserve (Fig. 3). Pitch actuators are slow, so pitch carries the steady reserve and the converter provides the fast response.

Reserve, stored energy and cost. Fig. 4 shows what the grid can draw. The reserve grows from 25.0 kW at 6 m/s to 154 kW at 11 m/s and 200 kW at rated wind speed. The kinetic energy that can be released down to the 0.5 pu speed limit grows from 0.60 MJ (equivalent inertia $H_{eq} = 0.30$ s) at 6 m/s to 4.34 MJ ($H_{eq} = 2.17$ s) at 10 m/s and 4.50 MJ ($H_{eq} = 2.25$ s) at 12 m/s. At 9 m/s one turbine releases 1.357 MJ between the deloaded speed and the MPPT speed ($H_{eq} = 0.68$ s) and 3.232 MJ down to the speed limit ($H_{eq} = 1.62$ s). The price is energy. For an illustrative Indian onshore site (Weibull $k = 2$, $c = 7.5$ m/s, Fig. 5) a permanent 10 percent deload cuts the annual energy from 4,725 MWh to 4,253 MWh (capacity factor 27.0 to 24.3 percent), a loss of 473 MWh or INR 16.5 lakh per year at 3.5 INR/kWh. This is an upper bound, because the reserve need only be held in low-inertia hours.

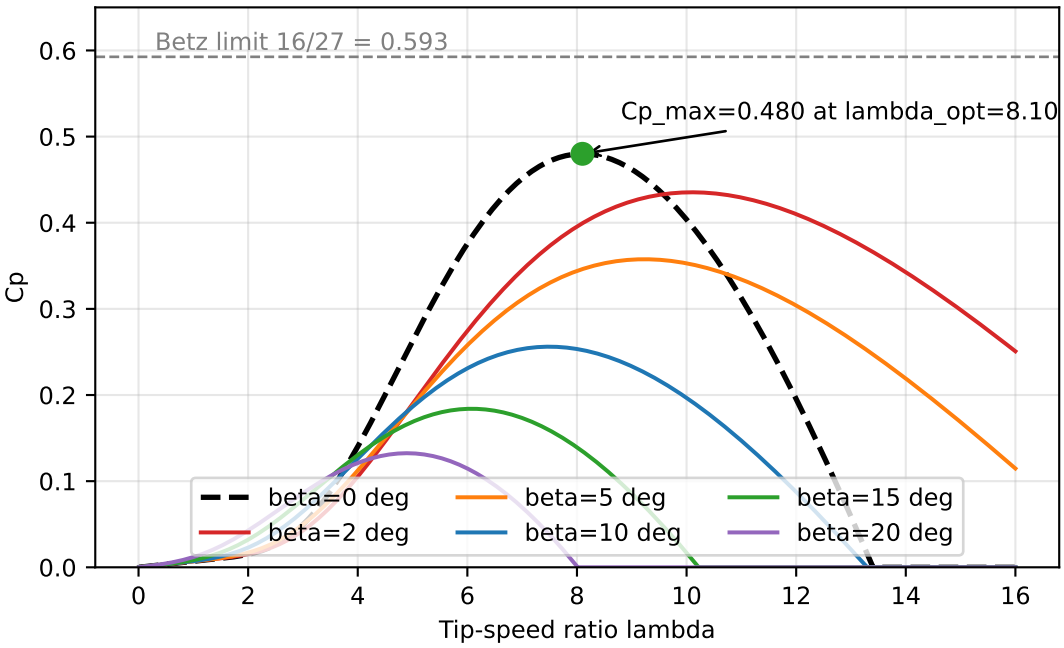


Figure 1: C_p versus TSR for several pitch angles, with the optimal and deloaded points.

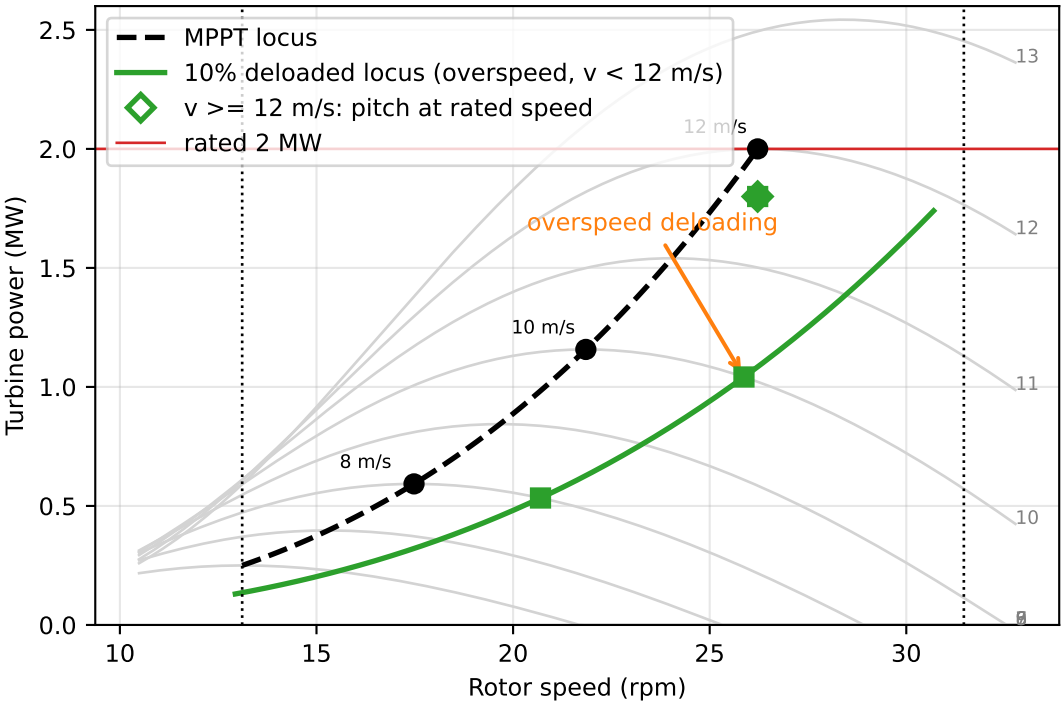


Figure 2: Turbine power versus rotor speed, MPPT line and overspeed deloaded line.

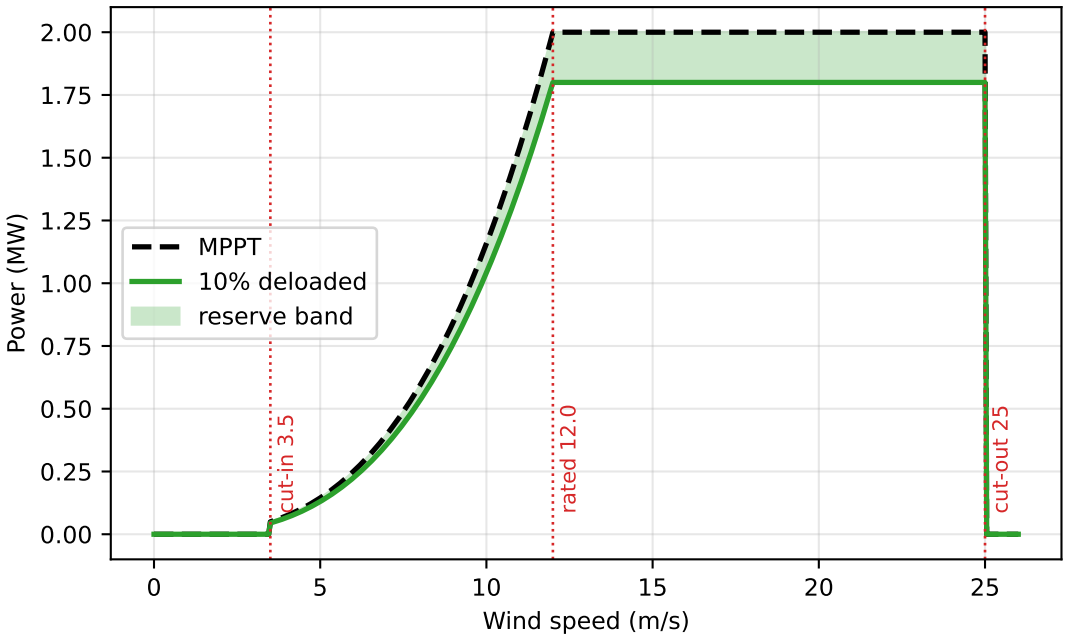


Figure 3: Power curve of the 2 MW turbine.

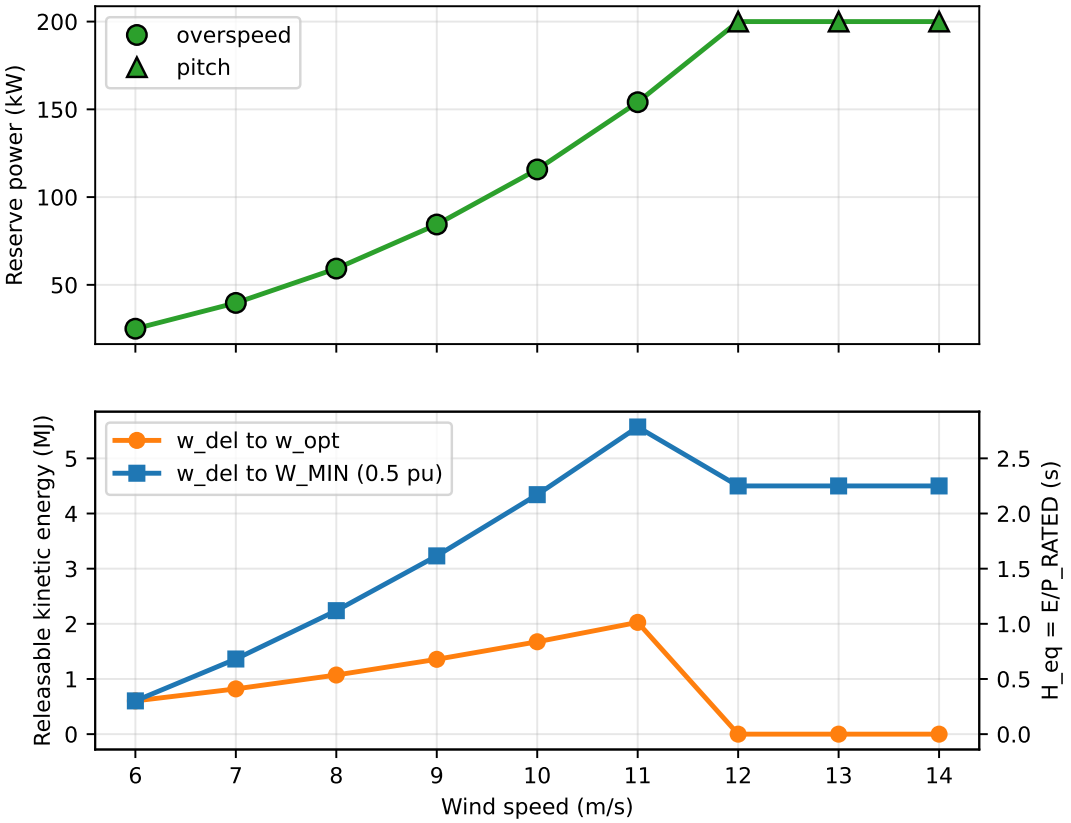


Figure 4: Power reserve and releasable kinetic energy versus wind speed.

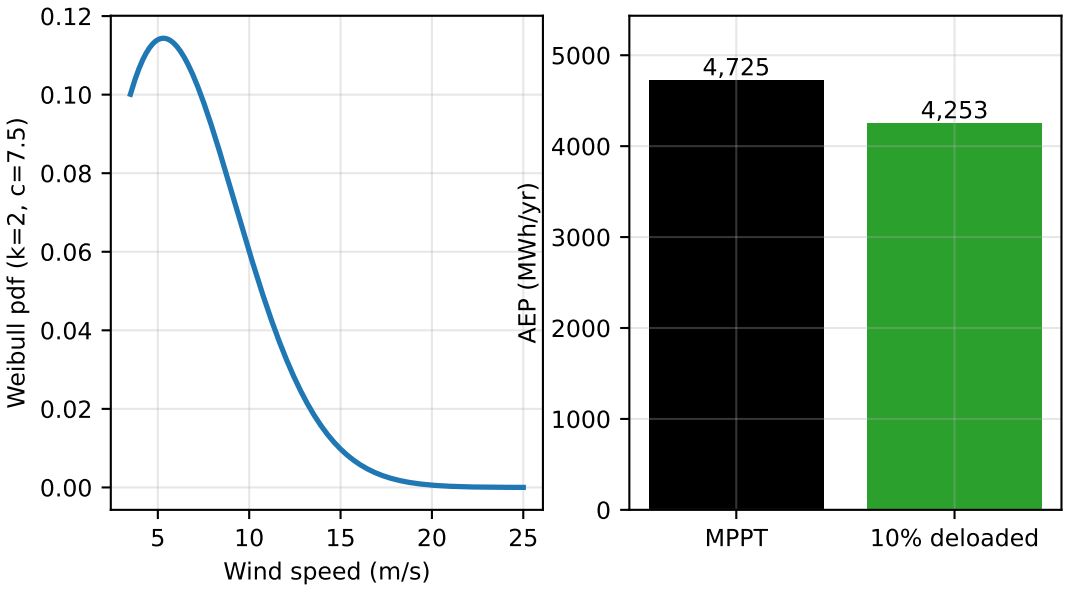


Figure 5: Annual energy at an illustrative Indian onshore site (Weibull $k = 2$, $c = 7.5$ m/s): MPPT versus 10 percent deloaded.

5.2 Frequency support

The cost of grid-following wind, and support without a reserve. Fig. 6 and Table 3 show a 50 MW (5 percent) load step at $t = 1$ s with 40 percent of the load supplied by wind at 9 m/s. Replacing synchronous generation by grid-following MPPT wind (C1) cuts the system inertia constant from 5.88 s (C0) to 3.76 s: the nadir falls from 49.771 to 49.659 Hz and the 500 ms RoCoF rises from 0.198 to 0.305 Hz/s. The MPPT turbine does not react to the event at all, its rotor stays at 0.750 pu. Adding synthetic inertia and droop to the MPPT turbine (C2) lowers the 500 ms RoCoF to 0.122 Hz/s, but the support is borrowed from the rotor alone. The rotor slows from 0.750 to 0.701 pu, leaves the C_p optimum, and aerodynamic power falls (Fig. 7). The frequency therefore keeps sagging to a late minimum of 49.815 Hz at 11.2 s after the step and settles at −0.160 Hz, worse than no support at all (C1, −0.148 Hz). This is the secondary-dip problem of non-deloaded turbines [6], appearing here as a prolonged sag.

Table 3: Frequency metrics after the 50 MW load step (40 percent wind, 9 m/s). t_{nadir} is measured from the step; the last column is the extra wind energy delivered in the 10 s after the step.

Case	Nadir (Hz)	t_{nadir} (s)	RoCoF inst. (Hz/s)	RoCoF 500 ms (Hz/s)	Δf at 40 s (Hz)	min ω_r (pu)	Extra energy (MJ)
C0 all-synchronous	49.771	2.14	0.213	0.198	−0.097	n/a	0
C1 GFL, MPPT	49.659	2.09	0.332	0.305	−0.148	0.750	0
C2 GFL+SI+droop, no reserve	49.815	11.20	0.332	0.122	−0.160	0.701	167
C3 GFL+SI+droop, 10% reserve	49.906	2.07	0.332	0.115	−0.070	0.831	317
C4 GFM-VSM, 10% reserve	49.906	2.03	0.162	0.116	−0.070	0.831	317

What the reserve does, and what grid-forming adds. With the 10 percent overspeed reserve (C3, C4) the rotor slows only from 0.888 to 0.831 pu and stays above the MPPT speed of 0.750 pu. Slowing therefore raises P_{aero} along the overspeed side of Fig. 2, as explained with Eq. (6): the plant delivers about 37 MW extra (peak 36.9 MW in C3 and 37.2 MW in C4), 317 MJ in 10 s. The nadir is 49.906 Hz, inside the IEGC band of 49.90 to 50.05 Hz, the steady-state deviation is −0.070 Hz and there is no secondary dip. GFM (C4) and GFL with the same reserve (C3) reach practically the same nadir (49.906 Hz in both), because the reserve sets the nadir. What GFM adds is at the step. Its instantaneous RoCoF is half that of C3 (0.162 against 0.332 Hz/s; centre-of-inertia value 0.148 against

0.268 Hz/s), because it acts as a voltage source and needs no frequency measurement, and it adds real inertia: H_{sys} rises from 3.76 to 8.51 s when the virtual inertia of the GFM plant is counted. Over a 500 ms window the two are similar (about 0.116 Hz/s), because our GFL frequency-measurement lag of 50 ms is optimistic. Step 2 will quantify realistic measurement windows. The decisive GFM advantage is weak-grid stability (Task C below).

Penetration and weak connection. Fig. 9 sweeps the wind share from 10 to 60 percent. The GFL-MPPT nadir worsens from 49.751 to 49.547 Hz, while the GFM nadir improves from 49.811 to 49.928 Hz, because every added GFM turbine adds both virtual inertia and reserve. The support-without-reserve case C2 is not monotonic (49.807 Hz at 10 percent, 49.827 Hz at 30 percent, 49.685 Hz at 60 percent) because it has no reserve to draw on. In Fig. 10 the connection of the GFM plant is weakened from $X_{23} = 0.1$ to 0.8 pu, so that the SCR at the wind bus falls from 3.01 to 1.00. The nadir stays at 49.906 Hz, the 500 ms RoCoF rises modestly from 0.112 to 0.148 Hz/s, and the damping ratio of the synchronous-generator/VSM electromechanical mode rises from 0.250 to 0.600 (mode frequency 1.83 to 0.89 Hz).

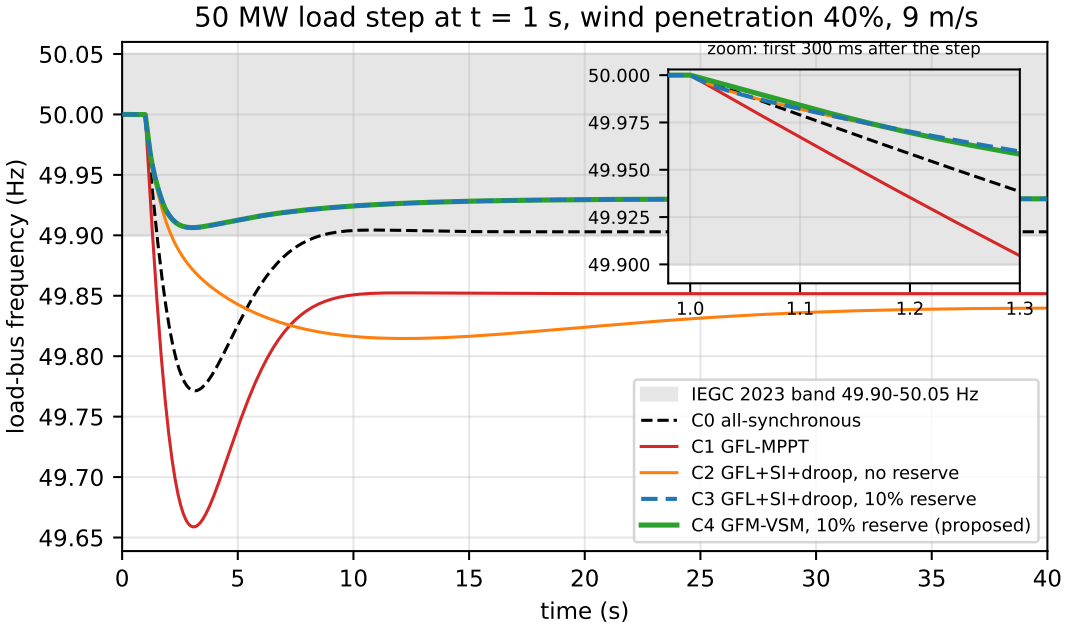


Figure 6: System frequency after the 50 MW load step for the cases C0 to C4.

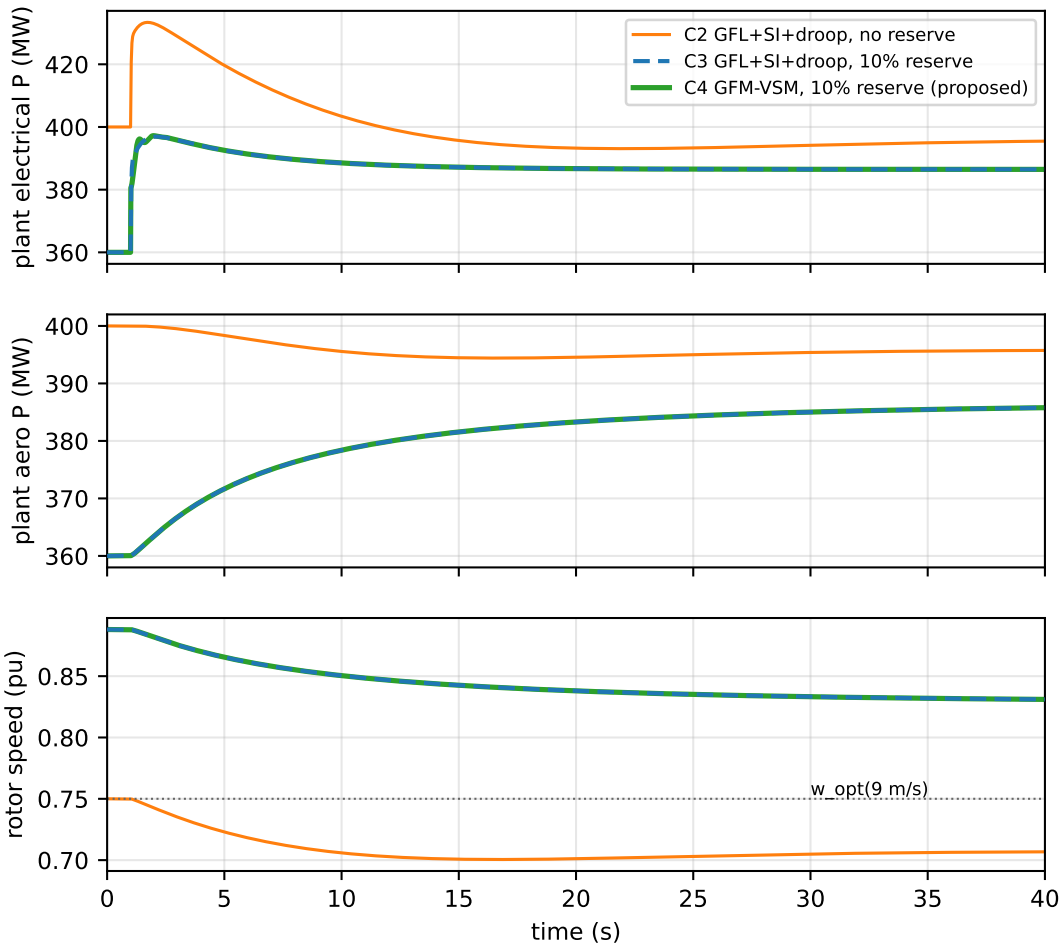


Figure 7: Wind turbine response during the event.

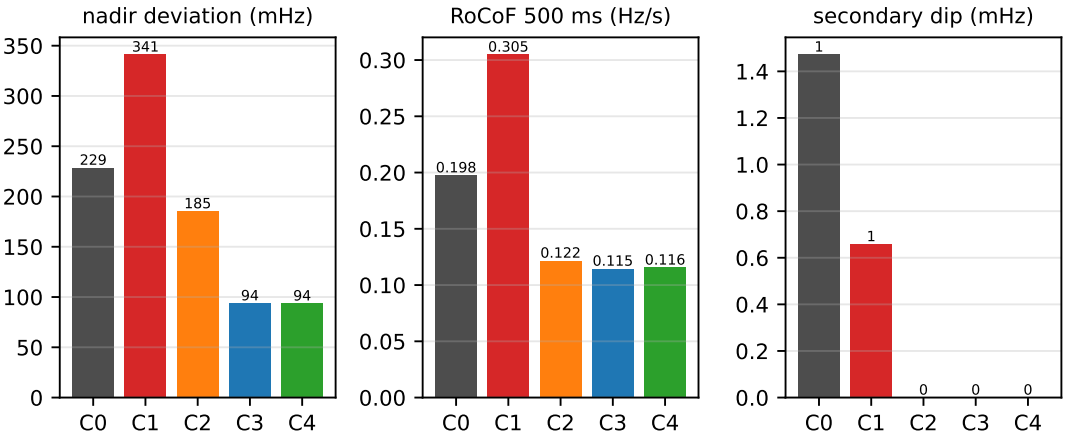


Figure 8: Frequency metrics (nadir, RoCoF) by case.

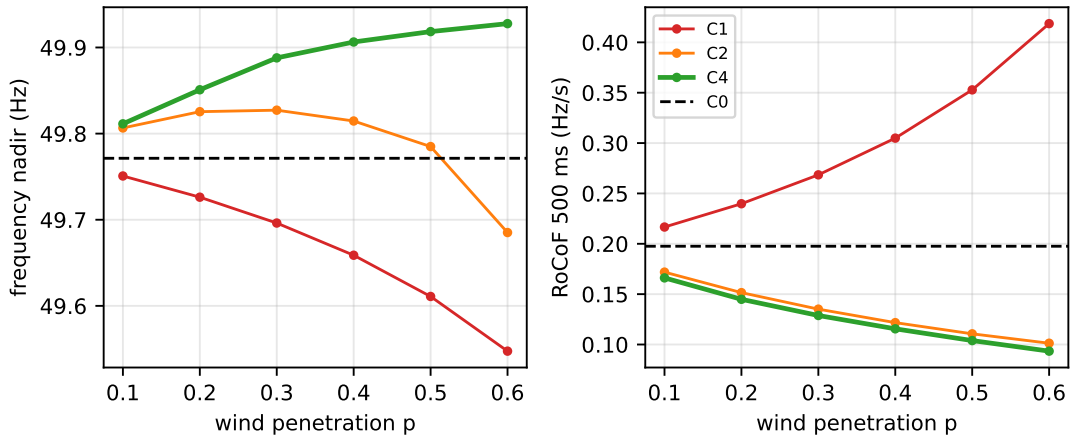


Figure 9: Sensitivity to wind penetration.

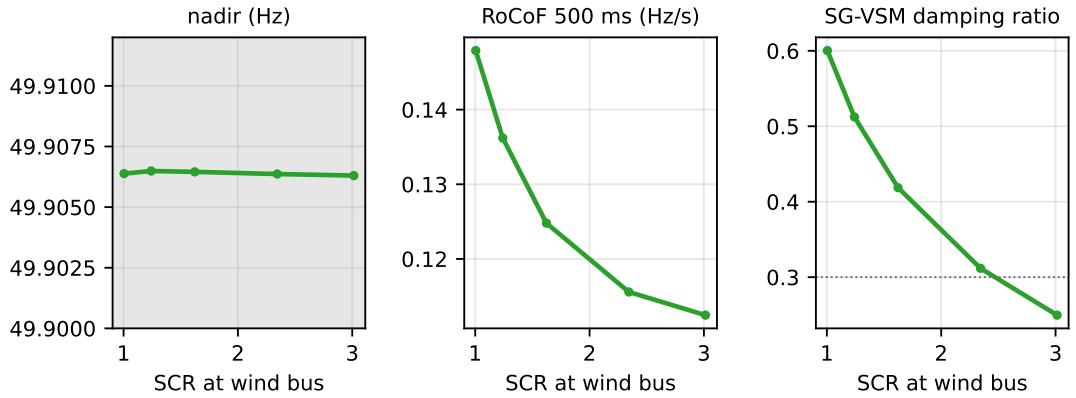


Figure 10: Effect of a weak connection on frequency support.

5.3 Weak-grid stability

Grid-following converter. A converter exports 1 pu to an infinite bus through $X_g = 1/\text{SCR}$ in a quasi-static phasor model, with the SCR swept from 10 down to 1.0. The GFL converter uses an SRF-PLL (damping 0.707, bandwidth 10, 20 or 40 Hz), an AC-voltage PI loop holding $|v_{pcc}| = 1$ and a 2 ms current-loop lag; the GFM converter is a VSM with $X_v = 0.15$ pu, $D = 20$ pu and $H = 2$ or 5 s. Fig. 11(a) shows that the GFL converter loses small-signal stability before it runs out of equilibria: an oscillatory eigenvalue pair crosses into the right half plane at a critical SCR of 1.367, 1.332 and 1.298 for the 10, 20 and 40 Hz PLL, against static limits of SCR 2.0 (unity power factor, $i_q = 0$) and 1.0 (with voltage support). Inside the IEEE Std 1204 weak-grid band (SCR below 3) the least-damped oscillatory mode of the 10 Hz PLL falls from a damping ratio of 0.702 at SCR 10 to 0.195 at SCR 1.5. Below SCR 2 a coupled PLL and voltage-loop oscillation of about 5 to 7 Hz takes over and becomes unstable.

What this model does not show. In this quasi-static model the PLL bandwidth shifts the GFL critical SCR only slightly (1.37, 1.33 and 1.30, a faster PLL being marginally better). The well-known penalty of a fast PLL needs the electromagnetic current-loop and network dynamics that this model omits; these are planned for Step 2, and we do not claim the penalty here.

Grid-forming converter and the duality. Fig. 11(b) and the eigenvalue loci of Fig. 12 show that the VSM stays stable at every SCR down to its static limit (last equilibrium at SCR 1.01) for both $H = 2$ s and $H = 5$ s, and that its swing-mode damping rises as the grid weakens: from 0.090 at

SCR 10 to 0.194 at SCR 1.5 for $H = 5$ s (0.142 to 0.307 for $H = 2$ s). This is the duality of Li, Gu and Green [5]: grid-following is robust in strong grids and fragile in weak ones, grid-forming is the opposite, and its weak spot is a lightly damped swing mode in strong grids ($\zeta \approx 0.09$ at SCR 10 for $H = 5$ s), which a larger damping gain or a virtual-impedance damper addresses. The result matters for India because large renewable complexes, such as those in western Rajasthan, feed long EHV corridors far from synchronous generation, which is the low-SCR regime. No substation-specific SCR is asserted here; site values must come from CEA and CTUIL system studies. The caveats are those of the model: a quasi-static network without line dynamics, no current-loop PI detail, no converter current limit, and a constant internal voltage of the VSM in the small-signal analysis.

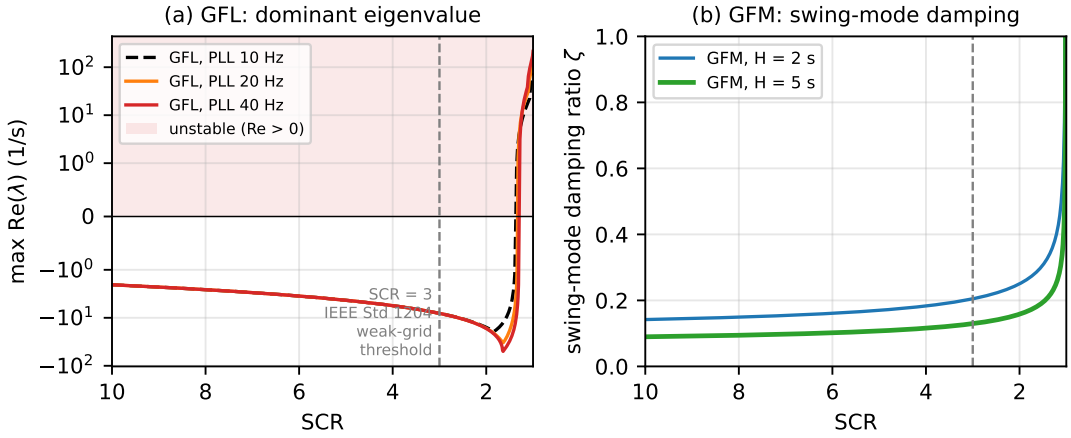


Figure 11: GFL versus GFM stability margin over the SCR sweep.

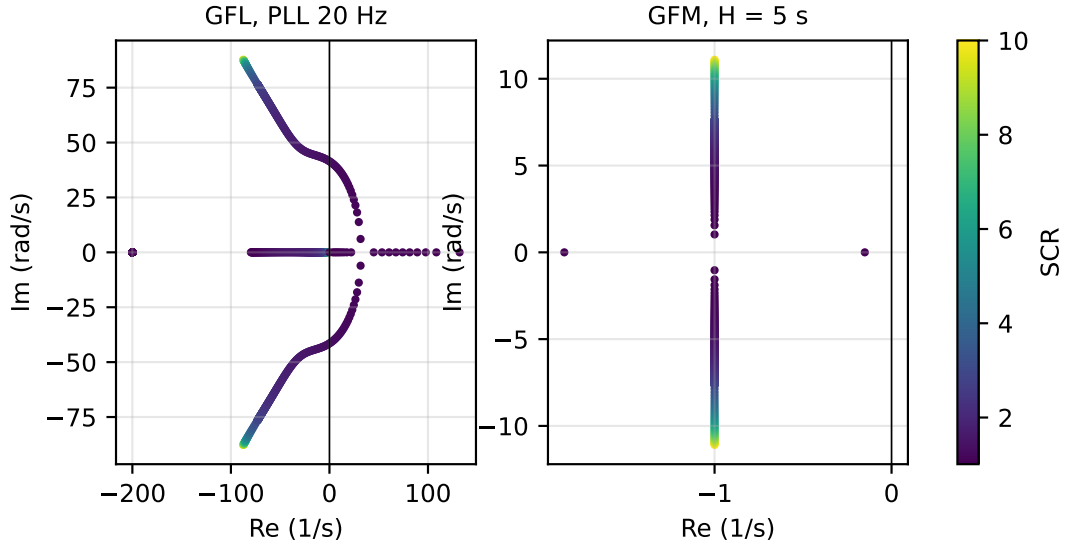


Figure 12: Eigenvalue loci over the SCR sweep.

6 Verification

Every model was checked by scripted gates against analytic results or invariants (tolerances are those coded in the simulation scripts). All 16 gates of Tasks A, B and C pass (Table 4); none failed and none is missing.

Table 4: Verification gates of Tasks A, B and C.

Gate	What it checks	Value	Status
GA1	$C_{p,max}$ is below the Betz limit	$0.4800 < 16/27 = 0.5926$	PASS
GA2	λ_{opt} and $C_{p,max}$ lie in the band expected for this rotor (7.5 to 8.7; 0.47 to 0.49)	$\lambda_{opt} = 8.100$, $C_{p,max} = 0.4800$	PASS
GA3	Rated power at rated wind speed (tolerance 1 percent)	$P_{avail}(12\ \text{m/s}) = 2000$ kW	PASS
GA4	P_{aero} re-evaluated at each deloaded operating point reproduces P_{del} (6 to 14 m/s)	maximum relative error 6.5×10^{-14} (limit 10^{-3})	PASS
GA5	Deloaded speed is on the overspeed side, $\omega_{del} > \omega_{opt}$, at every overspeed wind speed	6 to 11 m/s, e.g. 23.28 against 19.67 rpm at 9 m/s	PASS
GA6	Annual-energy loss equals the 10 percent deload target (tolerance 0.5 percentage points)	loss 10.00 percent	PASS
GB1	Pre-event steady state: no drift before the step	$\Delta f = 0$ in all five cases; initial residual below 1.5×10^{-17}	PASS
GB2	Centre-of-inertia RoCoF after the step against the analytic $\Delta P\, f_0/(2H_{sys})$ (limit 15 percent)	error 0.19 percent (C0), 0.24 percent (C1), 0.93 percent (C4)	PASS
GB3	Steady-state deviation of C1 against the analytic value (limit 5 percent)	-0.14826 Hz analytic, -0.14826 Hz simulated; relative error 1.1×10^{-10}	PASS
GB4	Energy bookkeeping $\int (P_{aero} - P_e)\, dt = \Delta E_k$ for C2, C3, C4 (limit 1 percent)	relative errors 4.1×10^{-12}, 6.7×10^{-12}, 3.1×10^{-12}	PASS
GB5	The C4 rotor stays on the overspeed side (no stall)	minimum $\omega_r = 0.8311$ pu against the limit 0.7400 pu	PASS
GB6	C4 beats C1 and C2 on nadir, and beats C1 on RoCoF	nadir 49.906 against 49.659 (C1) and 49.815 Hz (C2); RoCoF at 500 ms 0.116 against 0.305 Hz/s (C1)	PASS
GC1	GFL static limit (voltage loop off, $i_q = 0$): last SCR with an equilibrium against the theory $2P_0$ (limit 2 percent)	numeric 2.000, theory 2.000, error 0.00 percent	PASS
GC2	GFM swing mode at SCR 5: numerical eigenvalue against closed form (limit 2 percent)	$\omega_n = 14.713$ and 9.305 rad/s, $\zeta = 0.1699$ and 0.1075 for $H = 2$ and 5 s; errors below 10^{-9}	PASS
GC3	GFL stable in a strong grid	at SCR 10 the maximum real part is -2.05 1/s for the 10, 20 and 40 Hz PLL	PASS
GC4	Trends: GFL damping lowest at SCR 1.5 and every GFL variant reaches the right half plane; GFM damping rises as SCR falls and is monotone for $H = 5$ s	GFL 10 Hz: $\zeta = 0.702,\ 0.646,\ 0.195$ at SCR 10, 3, 1.5; GFM $H = 5$ s: 0.090 at SCR 10, 0.158 at SCR 2	PASS

7 Plan for Step 2 and Step 3

Table 5: Work plan.

Weeks	Task
Step 2, 1–2	Detailed PMSG dq model: machine-side and grid-side converters, DC link
Step 2, 3–4	Dual-port GFM control after Lyu and Gross [1]
Step 2, 5	Current limiting and fault ride-through
Step 3, 6–7	IEEE 9-bus test system; wind-speed turbulence
Step 3, 8–9	Comparison with Indian grid-code requirements [14, 13]
Step 3, 10	Final report and presentation

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